

Complex Analysis

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Contents

Chapter 1

Complex Numbers

A complex number is an expression of the form $z = x + iy$, where x and y are real numbers. The component x is called the real part of z , and y is the imaginary part of z . We will denote these by

$$\begin{aligned}x &= \operatorname{Re} z, \\y &= \operatorname{Im} z.\end{aligned}$$

The set of complex numbers forms the complex plane, which we denote by $\mathbb{C}$. We denote the set of real numbers by $\mathbb{R}$, and we think of the real numbers as being a subset of the complex plane, consisting of the complex numbers with imaginary part equal to zero.

The correspondence

$$z = x + iy \longleftrightarrow (x, y)$$

is a one-to-one correspondence between complex numbers and points (or vectors) in the Euclidean plane $\mathbb{R}^2$. The real numbers correspond to the x -axis in the Euclidean plane. The complex numbers of the form iy are called purely imaginary numbers. They form the imaginary axis $i\mathbb{R}$ in the complex plane, which corresponds to the y -axis in the Euclidean plane.

1.0.1 Basic Definitions and Algebra

- A complex number has the form $z = a + ib$ where a, b are real numbers
- a is the **real part**: $\operatorname{Re}(z) = a$
- b is the **imaginary part**: $\operatorname{Im}(z) = b$
- A number is **purely imaginary** if $a = 0$
- A number is **real** if $b = 0$
- Zero is the only number that is both real and purely imaginary

1.0.2 Arithmetic Operations

1.0.2.1 Addition

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

1.0.2.2 Multiplication

$$(a + ib)(c + id) = (ac - bd) + i(ad + bc)$$

1.0.2.3 Division

$$\frac{a + ib}{c + id} = \frac{(a + ib)(c - id)}{(c + id)(c - id)} = \frac{ac + bd}{c^2 + d^2} + i \frac{bc - ad}{c^2 + d^2}$$

1.0.3 Complex Conjugate

- The conjugate of $z = a + ib$ is $\bar{z} = a - ib$
- Properties:
 - $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$
 - $\overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$
 - $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$
 - z is real if and only if $z = \bar{z}$
 - z is purely imaginary if and only if $z = -\bar{z}$
- Expressing real and imaginary parts using conjugate:
 - $\operatorname{Re}(z) = \frac{z + \bar{z}}{2}$
 - $\operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$

1.0.4 Absolute Value (Modulus)

- The absolute value or modulus of $z = a + ib$ is:
 - $|z| = \sqrt{a^2 + b^2} = \sqrt{z\bar{z}}$
- Properties:
 - $|z| \geq 0$ for all z , with equality if and only if $z = 0$
 - $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$
 - $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ for $z_2 \neq 0$
 - $|z_1 + z_2| \leq |z_1| + |z_2|$ (Triangle inequality)
 - $||z_1| - |z_2|| \leq |z_1 - z_2|$

1.0.5 Important Inequalities

1. **Triangle Inequality:** $|z_1 + z_2| \leq |z_1| + |z_2|$

- Equality holds if and only if $\frac{z_1}{z_2} > 0$ (i.e., the ratio is positive real)

2. $-|z| \leq \operatorname{Re}(z) \leq |z|$

- Equality on right when z is positive real

3. $-|z| \leq \operatorname{Im}(z) \leq |z|$

4. **Cauchy's Inequality:** $|a_1 b_1 + a_2 b_2 + \dots + a_n b_n|^2 \leq (|a_1|^2 + \dots + |a_n|^2)(|b_1|^2 + \dots + |b_n|^2)$

- Equality holds if and only if the a_i are proportional to the b_i

1.1 Geometric Representation

1.1.1 The Complex Plane

- A complex number $z = a + ib$ corresponds to the point (a, b) in the plane
- The x -axis is called the **real axis**
- The y -axis is called the **imaginary axis**
- The plane is called the **complex plane**

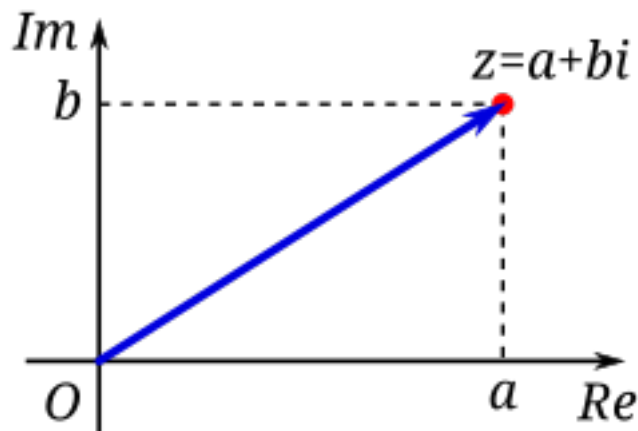


Figure 1.1: Image Source: Wikimedia

1.1.2 Vector Interpretation

- Complex numbers can be represented as vectors from the origin
- Vector addition corresponds to complex addition
- The length of the vector is $|z|$
- The distance between points z_1 and z_2 is $|z_1 - z_2|$

1.1.3 Polar Form

- A complex number can be written in **polar form**:
 - $z = r(\cos \theta + i \sin \theta)$ where $r = |z|$ and $\theta = \arg(z)$
 - r is the modulus: $r = |z| = \sqrt{a^2 + b^2}$
 - θ is the **argument** or **amplitude**: $\theta = \arg(z)$
- Properties of polar form:
 - For $z_1 = r_1(\cos \theta_1 + i \sin \theta_1)$ and $z_2 = r_2(\cos \theta_2 + i \sin \theta_2)$:
 - $z_1 z_2 = r_1 r_2 [\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$
 - $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$
 - $\arg\left(\frac{z_1}{z_2}\right) = \arg(z_1) - \arg(z_2)$

1.1.4 De Moivre's Formula

- For $z = r(\cos \theta + i \sin \theta)$:
 - $z^n = r^n(\cos n\theta + i \sin n\theta)$
- For $r = 1$:
 - $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

1.1.5 Roots of Complex Numbers

- The n -th roots of $z = r(\cos \theta + i \sin \theta)$ are:
 - $\sqrt[n]{z} = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right]$ for $k = 0, 1, 2, \dots, n-1$
- There are n distinct n -th roots of any non-zero complex number
- These roots have equal moduli and are equally spaced around a circle
- For $z = 1$, the roots are called the **n -th roots of unity**

1.1.6 Equations in the Complex Plane

- Circle with center a and radius r : $|z - a| = r$
 - In algebraic form: $(z - a)(\bar{z} - \bar{a}) = r^2$
- Straight line: $z = a + bt$ where t is a real parameter
 - Two lines are parallel if b' is a real multiple of b
 - Lines are orthogonal if $\frac{b'}{b}$ is purely imaginary
 - The angle between lines is $\arg \frac{b'}{b}$

1.1.7 Stereographic Projection and the Riemann Sphere

The extended complex plane is the complex plane together with the point at infinity. We denote it by $C^* = C \cup \{\infty\}$. - The extended complex plane includes ∞ and can be represented on a unit sphere - Under stereographic projection, the point (x_1, x_2, x_3) on the sphere corresponds to: $z = \frac{x_1 + ix_2}{1 - x_3}$

- The north pole $(0, 0, 1)$ corresponds to ∞
- Properties:
 - Circles on the sphere correspond to circles or lines in the plane
 - The distance formula: $d(z, z') = \frac{2|z - z'|}{\sqrt{(1 + |z|^2)(1 + |z'|^2)}}$
 - The sphere is often called the **Riemann sphere**

1.1.8 Key Formulas and Results

1. $z\bar{z} = |z|^2 = a^2 + b^2$
2. $\frac{1}{z} = \frac{\bar{z}}{|z|^2} = \frac{a - ib}{a^2 + b^2}$
3. $z^n = r^n(\cos n\theta + i \sin n\theta)$
4. $\sqrt[n]{z} = \sqrt[n]{r} \left[\cos\left(\frac{\theta + 2\pi k}{n}\right) + i \sin\left(\frac{\theta + 2\pi k}{n}\right) \right]$ where $k = 0, 1, \dots, n - 1$
5. Triangle inequality: $|z_1 + z_2| \leq |z_1| + |z_2|$
6. Cauchy's inequality: $\left| \sum_{i=1}^n a_i \bar{b}_i \right|^2 \leq \left(\sum_{i=1}^n |a_i|^2 \right) \left(\sum_{i=1}^n |b_i|^2 \right)$
7. The equation of a circle: $|z - a| = r$
8. The roots of unity: ω^k where $\omega = \cos \frac{2\pi}{n} + i \sin \frac{2\pi}{n}$ and $k = 0, 1, \dots, n - 1$

Chapter 2

Complex Functions

Chapter 3

Exercise

1. Find the values of

$$(1+2i)^3, \quad \frac{5}{-3+4i}, \quad \left(\frac{2+i}{3-2i}\right)^2, \quad (1+i)^n + (1-i)^n.$$

2. If $z = x + iy$ (where x and y are real numbers), find the real and imaginary parts of:

$$z^4, \quad \frac{1}{z}, \quad \frac{z-1}{z+1}, \quad \frac{1}{z^2}.$$

3. Show that:

$$\left(\frac{-1 \pm i\sqrt{3}}{2}\right)^3 = 1 \quad \text{and} \quad \left(\frac{\pm 1 \pm i\sqrt{3}}{2}\right)^6 = 1$$

for all combinations of signs.

Chapter 4

Green functions

Let G be a region and $a \in G$. Green functions G with singularity at a , the unique function $g_a : G \setminus \{a\}$

1. g_a is harmonic on $G \setminus \{a\}$.
2. $g_a(z) + \log(|z - a|)$ is harmonic¹ on an open neighbourhood of a .
3. $\lim_{z \rightarrow w} g_a(z) = 0 \quad \forall w \in \partial_\infty G$.

∴ {remark} $g_a(z) > 0$ for all $z \in G \setminus \{a\}$ because $g_a(z) + \log(|z - a|)$ has continuous extension to u and $\lim_{z \rightarrow w} g_a(z) = -\infty$ ∴

We have $g_a(z) < 0$ on $|z - a| = r$ for r small enough (then there exist R such that its positive $\forall r < R$) Given $r < R$ such that z_0 is $|z - a| = r$. Apply max principle on $G \setminus \{z : |z - a| \leq r\}$. Since $g_a(z) \geq 0$ or ∴

¹has an exterior to u . ($a \in u$), u is open in G .

Chapter 5

Riemann Surfaces.

Definition 5.1 (Riemann surface). A **Riemann surface** is a second-countable, Hausdorff topological space X equipped with a **complex atlas** $\mathcal{A} = \{(U_\alpha, \varphi_\alpha)\}$ satisfying the following properties:

1. Each $U_\alpha \subseteq X$ is open.
2. Each map $\varphi_\alpha : U_\alpha \rightarrow V_\alpha$ is a homeomorphism onto an open set $V_\alpha \subseteq \mathbb{C}$.
3. The collection $\{U_\alpha\}$ covers X .
4. For every pair of charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) with $U_\alpha \cap U_\beta \neq \emptyset$, the **transition map**

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is a **biholomorphism**.

Remark. Terminology: -“holomorphic functions” are functions which are complex differentiable on an open set in the complex plane. -A “biholomorphic map” is a holomorphic bijection whose inverse is holomorphic.

Definition 5.2. Let R_1 and R_2 be Riemann surfaces. A map $f : R_1 \rightarrow R_2$ is said to be **holomorphic** if for every pair of charts (U, φ) on R_1 , (V, ψ) on R_2 , the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic on its domain (which may be empty).

Suppose $(U, \varphi), (U', \varphi')$ are charts on R_1 , and $(V, \psi), (V', \psi')$ are charts on R_2 . On the overlap where all maps are defined, we have:

- the transition maps $\varphi' \circ \varphi^{-1}, \psi' \circ \psi^{-1}$ are **biholomorphisms** (by condition (5) in the definition of a complex atlas)

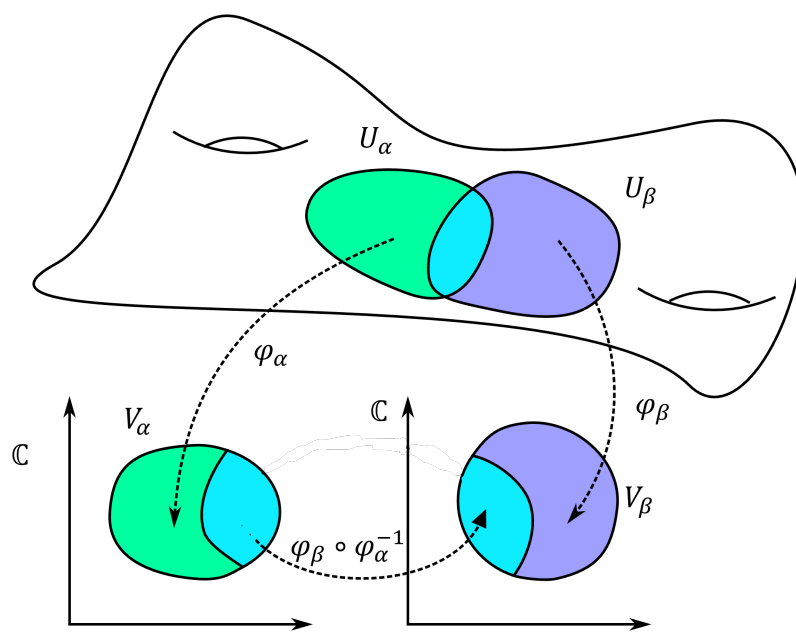


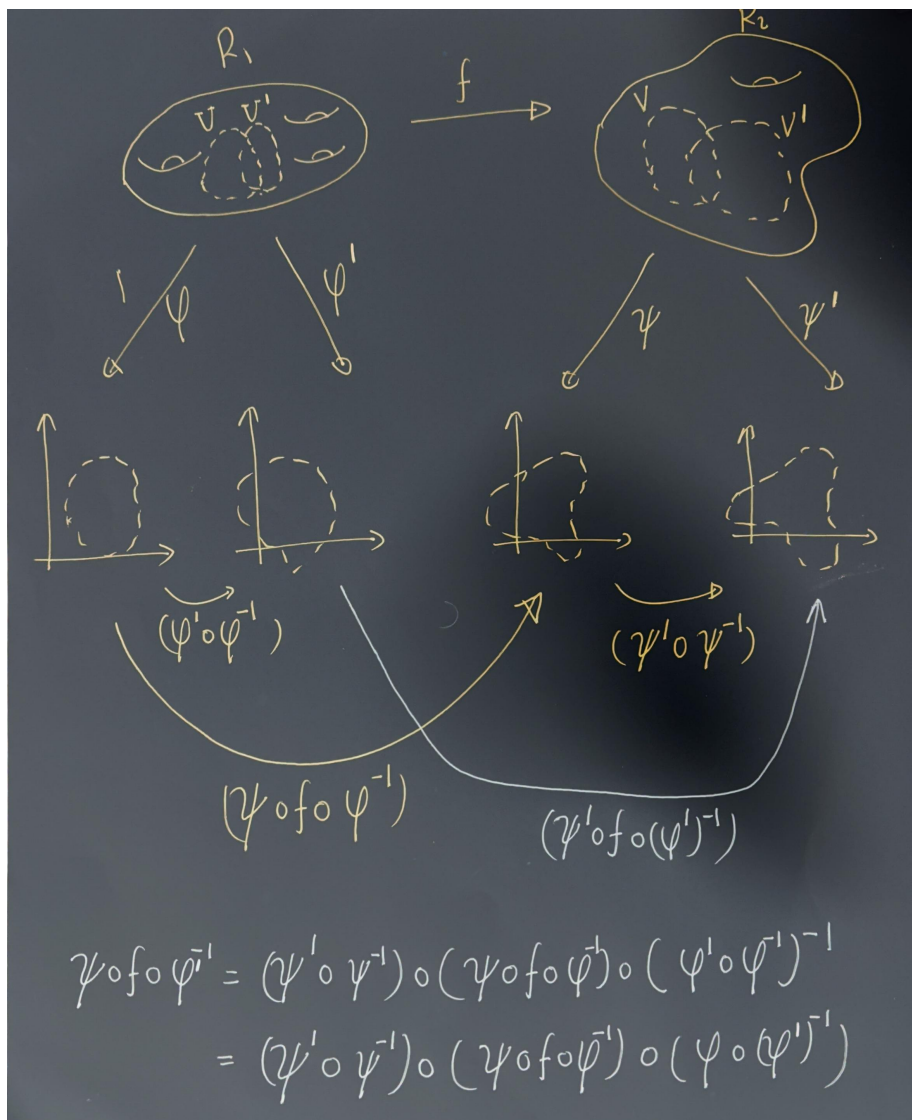
Figure 5.1: Two coordinate charts on a manifold (Image Credit: Wikipedia)

- therefore

$$\psi' \circ f \circ (\varphi')^{-1} = (\psi' \circ \psi^{-1}) \circ (\psi \circ f \circ \varphi^{-1}) \circ (\varphi \circ (\varphi')^{-1})$$

is holomorphic if and only if $\psi \circ f \circ \varphi^{-1}$ is holomorphic.

Thus the notion of holomorphicity does **not** depend on the choice of charts. “



Exercise 5.1. Let R_1 and R_2 be Riemann surfaces, and let $f : R_1 \rightarrow R_2$ be a continuous map. Assume that for every point $p \in R_1$ there exist:

- a chart (U_p, φ_p) on R_1 with $p \in U_p$,
- a chart (V_p, ψ_p) on R_2 with $f(p) \in V_p$,

such that

$$\psi_p \circ f \circ \varphi_p^{-1}$$

is holomorphic (in the usual sense) on its domain, an open subset of $\mathbb{C}$.

Show that f is holomorphic (in the sense of the definition for maps between Riemann surfaces).

Solution: I will update it later.

Example 5.1. Define an equivalence relation on $\mathbb{C}$ by $z \sim w \iff w = z + 2\pi ik$ for some $k \in \mathbb{Z}$. Let $R = \mathbb{C}/\sim$ be the quotient topological space, with projection map $\pi : \mathbb{C} \rightarrow R$. We will define an atlas on R that makes it a Riemann surface. Define $U = \{z \in \mathbb{C} : -\pi < \text{Im } z < \pi\}$. Restrict to U for convenience. Let $W = \pi(U) \subset R$. Let $\sigma : W \rightarrow U$ be the unique right inverse of π on W , i.e. the unique map satisfying:

- $\pi \circ \sigma = \text{id}_W$,
- $\sigma(W) = U$.

Exercise 5.2. Show that for any $p \in W$ and any $q \in W$, the map $\sigma(q) - \sigma(p) : U \rightarrow \mathbb{C}$ is a **translation**, and in particular holomorphic.

5.1 The ann sphere

As a set, the **Riemann sphere** is $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$. To make this into a Riemann surface, we must specify a **topology** and an **atlas**.

A basis for the topology consists of:

1. Sets of the form $U \subset \mathbb{C}$, where U is open in $\mathbb{C}$ and $\infty \notin U$.
2. Sets of the form $\{\infty\} \cup (\mathbb{C} \setminus K)$, where $K \subset \mathbb{C}$ is compact.

Equivalently, a set $V \subset \widehat{\mathbb{C}}$ is open iff:

- If $p \in \mathbb{C}$, then there exists an open set $U \subset \mathbb{C}$ with $p \in U \subset V$.
- If $p = \infty$, then there exists a compact $K \subset \mathbb{C}$ such that $\{\infty\} \cup (\mathbb{C} \setminus K) \subset V$.

Exercise 5.3. Show that $\widehat{\mathbb{C}}$ is compact.